Literature Status: The Lorentz `lorentzck` Conjecture

Run `fa_banach_001`, lane 11

2026-07-18

Status

This is a lightweight literature-status packet. The Lorentz-space conjecture from Defant, Mastyłło, and Michels, arXiv:math/0006034, is already answered in later s-number literature by Hinrichs and by Hinrichs–Michels. This packet is not an original proof claim by the run.

Source Conjecture

The source paper is Andreas Defant, Mieczyslaw Mastyłło, and Carsten Michels, *Summing inclusion maps between symmetric sequence spaces*, arXiv:math/0006034; published in *Transactions of the American Mathematical Society* 354 (2002), 4473–4492.

Near the end of the paper, on page 19 of the locally rendered `source_paper.pdf` and after the corollary containing the displayed formula labelled `lorentzck`, the authors conjecture that

$$c_k\big(\mathrm{id} : \ell^n_{v_1,v_2} \to \ell^n_{u_1,u_2}\big) \asymp a_k\big(\mathrm{id} : \ell^n_{v_1,v_2} \to \ell^n_{u_1,u_2}\big) \asymp (n-k+1)^{1/u_1-1/v_1}$$

for every $1 < u_1 < v_1 < 2$ and every $1 \le u_2, v_2 \le \infty$. The source paper proves only special secondary-parameter ranges immediately before this sentence.

Supporting Literature

The approximation-number side is covered by Aicke Hinrichs, *Approximation Numbers of Identity Operators between Symmetric Sequence Spaces*, *Journal of Approximation Theory* 118 (2002), 305–315. Its abstract states that it proves asymptotic formulas for approximation quantities of identity operators between symmetric sequence spaces and gives applications to Lorentz and Orlicz sequence spaces, extending Defant–Mastyłło–Michels.

The Gelfand-number side is covered by Aicke Hinrichs and Carsten Michels, *Gelfand Numbers of Identity Operators Between Symmetric Sequence Spaces*, *Positivity* 10 (2006), 111–133. The introduction explicitly identifies the Defant–Mastyłło–Michels conjecture as formula (1.1), involving both a_k and c_k, and states that the paper proves that formula under its interpolation hypotheses. The paper also notes that the displayed formulas may be read with approximation numbers by replacing the Gelfand/Kolmogorov symbols with a_k.

For the present target, Example 7.5(ii) in the 2006 paper gives

$$c_k\big(\mathrm{id} : \ell^n_{q_1,q_2} \to \ell^n_{p_1,p_2}\big) \asymp (n-k+1)^{1/p_1-1/q_1}$$

for $0 < p_1 < q_1 < \infty$ and $0 < p_2, q_2 \le \infty$.

Identification

Set

$$p_1 = u_1, \qquad q_1 = v_1, \qquad p_2 = u_2, \qquad q_2 = v_2.$$

The supporting range $0 < p_1 < q_1 < \infty$, $0 < p_2, q_2 \leq \infty$ contains the source range $1 < u_1 < v_1 < 2$, $1 \leq u_2, v_2 \leq \infty$. Under this substitution, the exponent $1/p_1 - 1/q_1$ is exactly $1/u_1 - 1/v_1$, and the identity map is precisely

$$\mathrm{id} : \ell^n_{v_1,v_2} \to \ell^n_{u_1,u_2}.$$

Together with the approximation-number literature cited above, this answers the full displayed `lorentzck` conjecture from the source paper.

Search Evidence and Scope

Cheap run indexes showed no prior exact packet for arXiv:math/0006034. A bounded literature search for the source arXiv id, `lorentzck`, and the Hinrichs/Hinrichs–Michels titles located the two decisive supporting papers. The 2006 paper explicitly connects itself to the Defant–Mastyłło–Michels conjecture, so this packet is classified as `literature_already_answered`.

No novelty is claimed. The recorded result is only the already-known answer to the Lorentz `lorentzck` conjecture; unrelated questions from the source paper are outside this packet.

References

1. A. Defant, M. Mastyłło, and C. Michels, *Summing inclusion maps between symmetric sequence spaces*, Trans. Amer. Math. Soc. 354 (2002), 4473–4492. DOI: 10.1090/S0002-9947-02-03056-8.

2. A. Hinrichs, *Approximation Numbers of Identity Operators between Symmetric Sequence Spaces*, J. Approx. Theory 118 (2002), 305–315. DOI: 10.1006/jath.2002.3726.

3. A. Hinrichs and C. Michels, *Gelfand Numbers of Identity Operators Between Symmetric Sequence Spaces*, Positivity 10 (2006), 111–133. DOI: 10.1007/s11117-005-0022-1.